# POINTS OF A SITE WITH FINITE LIMITS ARE FUNCTORS COMMUTING WITH FINITE LIMITS AND COVERINGS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- The definition of a point of a site. Same chapter, Definition definition-point. Supplies the conditions shown to be equivalent to the two displayed ones.
- Stalks of representable sheaves. Same chapter, Lemma lemma-points-recover and Lemma lemma-point-pushforward-sheaf. Identify the functor with the composite of the representable functor, sheafification and the stalk functor.
- Directedness of the category of neighbourhoods. Same chapter, Lemma lemma-neighbourhoods-directed. Gives commutation with finite limits for a functor satisfying the two conditions.

*Source reference.* Original definition 00Y5: sources/stacks-sites/pinned-original-source.tex, source lines 8539–8539. Navigation pointer invoked at proof line 8539; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00Y6: sources/stacks-sites/pinned-original-source.tex, source lines 8541–8541. Navigation pointer invoked at proof line 8541; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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*Date*: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 00Y8: sources/stacks-sites/pinned-original-source.tex, source lines 8542–8542. Navigation pointer invoked at proof line 8542; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00YB: sources/stacks-sites/pinned-original-source.tex, source lines 8559–8559. Navigation pointer invoked at proof line 8559; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Proposition 0.1.** *Let $\mathcal{C}$ be a site. Assume that finite limits exist in $\mathcal{C}$. (I.e., $\mathcal{C}$ has fibre products, and a final object.) A point $p$ of such a site $\mathcal{C}$ is given by a functor $u : \mathcal{C} \to Sets$ such that*

(1) *$u$ commutes with finite limits, and*
(2) *if $\{U_i \to U\}$ is a covering, then $\coprod_i u(U_i) \to u(U)$ is surjective.*

*Proof.* Suppose first that $p$ is a point (Definition 00Y5) given by a functor $u$. Condition (2) is satisfied directly from the definition of a point. By Lemma 00Y6 we have $(h_U)_p = u(U)$. By Lemma 00Y8 we have $(h_U^\#)_p = (h_U)_p$. Thus we see that $u$ is equal to the composition of functors

$$\mathcal{C} \xrightarrow{h} PSh(\mathcal{C}) \xrightarrow{\#} Sh(\mathcal{C}) \xrightarrow{()_p} Sets$$

Each of these functors is left exact, and hence we see $u$ satisfies (1).

Conversely, suppose that $u$ satisfies (1) and (2). In this case we immediately see that $u$ satisfies the first two conditions of Definition 00Y5. And its stalk functor is exact, because it is a left adjoint by Lemma 00Y8 and it commutes with finite limits by Lemma 00YB. □